Table 79. UFBGA64 – Mechanical data

Symbol	millimeters[1]			inches[12]		
	Min	Typ	Max	Min	Typ	Max
A[2][3]	-	-	0.60	-	-	0.0236
A1[4]	0.05	-	-	0.0020	-	-
A2	-	0.43	-	-	0.0169	-
b[5]	0.23	0.28	0.33	0.0090	0.0110	0.0130
D[6]	5.00 BSC			0.1969 BSC		
D1	3.50 BSC			0.1378 BSC		
E	5.00 BSC			0.1969 BSC		
E1	3.50 BSC			0.1378 BSC		
e[9]	0.50 BSC			0.0197 BSC		
N[11]	64					
SD[12]	0.25 BSC			0.0098 BSC		
SE[12]	0.25 BSC			0.0098 BSC		
aaa	0.15			0.0059		
ccc	0.20			0.0079		
ddd	0.08			0.0031		
eee	0.15			0.0059		
fff	0.05			0.0020		

Notes:

1. Dimensioning and tolerancing schemes conform to ASME Y14.5M-2009 apart European projection.
2. UFBGA stands for ultra profile fine pitch ball grid array: 0.5 mm < A ≤ 0.65 mm / fine pitch e < 1.00 mm.
3. The profile height, A, is the distance from the seating plane to the highest point on the package. It is measured perpendicular to the seating plane.
4. A1 is defined as the distance from the seating plane to the lowest point on the package body.
5. Dimension b is measured at the maximum diameter of the terminal (ball) in a plane parallel to primary datum C.
6. BSC stands for BASIC dimensions. It corresponds to the nominal value and has no tolerance. For tolerances refer to form and position table. On the drawing these dimensions are framed.
7. Primary datum C is defined by the plane established by the contact points of three or more solder balls that support the device when it is placed on top of a planar surface.
8. The terminal (ball) A1 corner must be identified on the top surface of the package by using a corner chamfer, ink or metalized markings, or other feature of package body or